

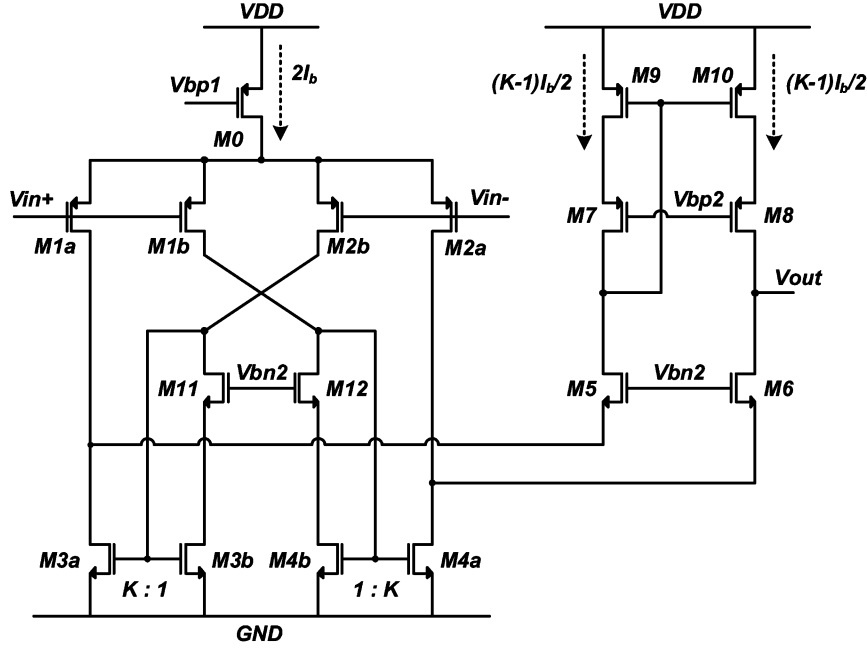


Fig. 2. The recycling folded cascode (RFC) amplifier.

We will refer to the modified FC as the recycling folded cascode (RFC), as we are reusing, or recycling, existing devices and currents to perform an additional task.

### III. RFC CHARACTERISTICS

The modifications presented in Section II provide the RFC with enhanced features over that of the FC. In order to present these enhancements quantitatively, all devices are assumed to operate in the saturation region following the simplified square-law drain current model given by

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_T)^2. \quad (1)$$

Here the symbols have their usual meanings of carrier mobility, gate oxide per unit area, device aspect ratio, gate-source voltage, and threshold voltage respectively. Also, the current gain  $K$  shown in Fig. 2 is selected to equal 3. This maintains the same power consumption for both the FC and RFC shown in Figs. 1 and 2. Note, however, that the current through  $M5 - M10$  is now a function of  $K$ ; for  $K \neq 3$ ,  $M5 - M10$  need to be scaled accordingly to maintain the same inversion level – equal power and area as the FC is only achieved at  $K = 3$ .

The analyses presented in Sections III-A–G are valid for both the single-ended and differential implementations of the amplifiers. Nonetheless, wherever appropriate further insight into the fully differential implementation will be provided.

#### A. Small Signal Transconductance

We first examine the amplifier's transconductance,  $G_m$ , by finding the short-circuit current at the output with respect to the input. The results for RFC and FC follow in (2) and (3).

$$G_{m_{RFC}} = g_{m_{1a}} (1 + K) \quad (2)$$

$$G_{m_{FC}} = g_{m_1}. \quad (3)$$

By taking into account that  $M1$  is twice the size of  $M1a$  and conducts twice the amount of current (i.e.,  $g_{m_1} = 2g_{m_{1a}}$ ), and substituting for the value of  $K$ , the transconductance of the RFC is demonstrated to be twice that of the FC for the same power consumption. This shows that the RFC has twice the gain-bandwidth (GBW) as that of the FC for the same power, and consequently twice the speed.

#### B. Low Frequency Gain

The low frequency gain of OTAs is frequently expressed as the product of the small signal transconductance,  $G_m$ , and the low frequency output impedance,  $R_o$ . It was demonstrated that  $G_{m_{RFC}} = 2G_{m_{FC}}$ , which results in a 6 dB gain enhancement for the same output impedance. However,  $R_{o_{RFC}}$  is also enhanced over  $R_{o_{FC}}$ . The expressions for  $R_{o_{RFC}}$  and  $R_{o_{FC}}$  are represented by (4) and (5), respectively.

$$R_{o_{RFC}} \cong g_{m_6} r_{ds6} (r_{ds2a} \parallel r_{ds4a}) \parallel g_{m_8} r_{ds8} r_{ds10} \quad (4)$$

$$R_{o_{FC}} \cong g_{m_6} r_{ds6} (r_{ds2} \parallel r_{ds4}) \parallel g_{m_8} r_{ds8} r_{ds10}. \quad (5)$$

The gain enhancement seen in  $R_{o_{RFC}}$  is attributed to the increased  $r_{ds}$  of  $M2a$  and  $M4a$ , as they conduct less current compared to their counterparts  $M2$  and  $M4$  of the FC. Therefore, an overall low frequency gain enhancement of 8–10 dB can be seen in the RFC compared to the FC.

This added gain has two fundamental benefits. First, static settling errors are reduced because of the increased gain. Second, the power supply rejection ratio (PSRR) performance of the RFC is improved over its FC counterpart. PSRR is defined as the injected supply noise gain with respect to the input signal gain. Both the FC and RFC have similar noise injections gains from either supply, but the loop gain of the RFC is higher and hence its PSRR performance is better. Moreover, the extended GBW of the RFC extends the improved PSRR performance to higher frequencies than the FC.